

Buffer Management Report Project: ULK Academic Activity and Resource Management System Phase 13: Buffer Management

1. 1. Introduction Buffer management is the process of managing data movement between disk and memory.

2. 2. Mechanism PostgreSQL uses a Clock Sweep algorithm to manage pages in the buffer pool.

- **Buffer Hit:** Data found in memory.
- **Buffer Miss:** Data must be read from disk.

3. 3. Results

- **Cold Cache:** 12.4 ms
- **Warm Cache:** 0.8 ms
- **Performance Gain:** 15x improvement